

B.Tech. Degree VI Semester Examination April 2013

ME 606 CAD/CAM (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) Write short notes on the various softwares used in CAD.
(b) Sketch the various levels of automation with mention to its flexibility.
(c) Differentiate an open loop system from a closed loop system.
(d) Write short notes on CAPP.
(e) What is the significance of APC?
(f) Discuss on special work holding devices in CNC machines.
(g) Differentiate PTP from CP systems.
(h) Write notes on AI systems.

PART B

(4 × 15 = 60)

- II. (a) Differentiate solid modelling from wire frame modelling . (5)
(b) With a suitable example, describe the basic steps of FEM process. (10)
OR
- III. (a) With neat sketches, explain the various transfer machines used. (10)
(b) Write notes on line balancing. (5)
- IV. (a) Describe the various position control system in CNC. (10)
(b) What are the various kinds of transducers used in CNC system? (5)
OR
- V. (a) Write notes on canned cycles. (5)
(b) Describe the various programming methods of CNC with mention to each of its merits and limitations. (10)
- VI. Describe the static and dynamic errors in CNC system. How it can be corrected? (15)
OR
- VII. (a) Compare a turning centre with a machining centre. (5)
(b) Describe the testing of CNC machine tools. (10)
- VIII. (a) Write notes on resolution, accuracy and repeatability of robots. (7)
(b) What are the various controls used in robots? (8)
OR
- IX. (a) Explain the concept of FMS in detail. What are the facilities used? (10)
(b) What are expert systems? (5)
